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SUPERSEDING
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Revision 1
May 29, 1956, and
Fed. Spec. TT-I-547
March 24, 1948

FEDERAL SPECIFICATION

INK, NUMBERING MACHINE, FOR PHOTOGRAPHIC NEGATIVES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—The numbering-machine ink covered by this specification is intended for use in marking on the uncoated side of photographic negatives. It is not suitable for other marking purposes.

1.2 **Classification.** — The numbering-machine ink shall be of one type and grade.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.**—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

GG–N–775 — Numbering Machines, Office Type, Hand Operated.

UU–P–63—Paper, Blotting.

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL–B–636—Boxes, Fiber, Solid (for Domestic Shipment).

PPP–B–585—Boxes; Wood, Wirebound.

PPP–B–591—Boxes, Fiberboard, Wood-Cleated.

PPP–B–601—Boxes, Wood, Cleated-Plywood.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

JAN–P–100—Packaging and Packing for Overseas Shipment—General Specification.

JAN–P–108—Boxes, Fiberboard, Corrugated and Solid (Military Type).

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JAN-P-127—Packaging and Packing for Overseas Shipment—Tape, Adhesive, Pressure-Sensitive, Water-Resistant.

MIL-B-10377—Boxes, Wood, Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Consolidated Classification Committee:

Consolidated Freight Classification Rules and Container Specifications.

(Application for copies of the Consolidated Freight Classification Rules and Container Specifications should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Materials.—The manufacturer is given wide range in the selection of materials and processes of manufacture, provided that the ink produced meets the requirements of this specification.

3.2 Properties.—The ink shall be suitable for use in numbering machines equipped with metal type and shall produce clear, legible, dense characters on the uncoated side of photographic negatives. Coagulation or precipitation shall not take place in the ink under ordinary working conditions or when the ink is stored for 6 months at temperatures in the range of 5° to 40°C. (41° to 104°F.).

3.3 Color.—The color shall be an intense black.

3.4 Density.—A layer of dried ink, applied by a numbering machine in the usual manner, shall have a density of not less than 2.0. (See 4.3.1.)

3.5 Drying time.—When the ink is applied with a numbering machine to the uncoated side of a dry photographic negative, the ink shall be dry to the touch in not more than 10 seconds and shall be completely dry in not more than 5 minutes when exposed to ambient conditions of 23°C. plus or minus 5°C. and 50 percent plus or minus 2 percent relative humidity. (See 4.3.2.)

3.6 Solubility.—The ink, when dried completely on the photographic negative, shall be insoluble in water up to 32°C. (90°F.) and the marking shall remain legible and opaque when tested as specified in 4.3.3.

3.7 Inked pad.—The ink shall remain moist on the ink pad of a numbering machine for at least 30 minutes and produce clear and legible markings. After an application of the ink has dried on the ink pad, it shall be possible to remoisten the pad with fresh ink and make the pad suitable for reuse. (See 4.3.4.)

3.8 Solvent.—A suitable ink solvent shall be available from the manufacturer for making erasures. Unless otherwise specified, a 1-ounce container of solvent shall be furnished for test purposes (see 6.4). The solvent shall be capable of removing inked images from photographic negatives immediately after application or as long as one year after application, without etching or otherwise damaging the film. The ink solvent and its vapors shall be nontoxic and shall not form inflammable mixtures with air.

3.9 Permanence. — Impressions shall not fade or reticulate nor become illegible or detached from the negatives when tested in accordance with 4.3.6.

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3.10 Stains.—The effects of the ink or solvent shall not cause any staining of the negative such as to cause a vignetting effect to appear in title area following 1 year of storage.

3.11 Containers.—The ink shall be furnished in 1-ounce circular-base glass bottles which shall be closed with a suitable screw cap. Each cap shall be fitted with a suitable polyethylene gasket to prevent evaporation of the ink. Each glass bottle shall be enclosed in an individual sealed chipboard container. (See 5.4.1.)

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, sampling shall be conducted in accordance with the provisions of Military Standard MIL–STD–105 and Appendix thereto. Inspection shall be made by the Government to determine compliance with the requirements of this specification. A 1-ounce sample of solvent for performing necessary tests (see 4.3.5) shall be furnished unless a larger sample is specified by the procuring officer. Sampling for inspection and tests shall be based on Inspection Level I and Acceptable Quality Level (approx.) =2.5 percent defective, in accordance with Military Standard MIL–STD–105.

4.2 Material for tests.

4.2.1 *Numbering machine.*—The numbering machine used for testing the ink shall conform to style 1 of Federal Specification GG–N–775, and shall be equipped with a clean inked pad.

4.2.2 *Film.*—Approximately 1 square foot or an equivalent area of unexposed photographic film shall be fixed, washed, dried, and cut into portions of suitable size. The density of the processed film shall not exceed 0.1.

4.3 Inspection and tests.

4.3.1 *Density.*—Moisten the ink pad in the numbering machine with the ink. Place a piece of negative, uncoated side up, on a single thickness of white blotting paper conforming to Federal Specification UU–P–63, and mark the negative with the numbering machine. Allow the ink to dry. Examine the markings for clarity and legibility. Measure the density of the dried ink by means of a densitometer, employing a spot not over 0.5 millimeter in diameter.

4.3.2 *Drying time.*—The drying time shall be conducted at 23°C. plus or minus 5°C. (73.5°F. plus or minus 9°F.) and 50 percent plus or minus 2 percent relative humidity. Mark a piece of negative with the numbering machine. Allow the work to dry for 10 seconds and immediately make one rub with a clean white cloth at ¼-ounce pressure. The marking shall be considered dry to the touch if there is no discoloration of the cloth by ink and the outlines of the markings are not blurred. Allow the marking to dry for an additional 4 minutes and 50 seconds and erase the marking with an art gum eraser under 8 ounces of pressure. Complete dryness of the marking will be indicated if it cannot be completely erased without the removal of a portion of the film surface.

4.3.3 *Solubility.*—Mark two pieces of negative and allow the markings to dry for not less than 1 hour. Immerse one of the pieces of marked negative for 1 hour in water having a temperature of 32°C. (90°F.). Remove the negative and compare the legibility of the markings with those on the negative which is not immersed. The markings on the immersed piece shall show no decrease in density.

4.3.4 *Inked pad.*—Moisten the ink pad of the numbering machine with the ink and expose to the air for 30 minutes. Immediately mark a piece of negative. Examine the markings for clarity and legibility. Allow the ink to dry on the pad. Remoisten with ink and determine by marking the negative if the pad is suitable for reuse. Should the pad be unsuitable for reuse, it shall be cause for rejection.

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4.3.5 *Solvent.*—Mark a negative and allow the markings to dry for 1 hour. Moisten a piece of clean cotton cloth with solvent and rub the dried markings with the moistened cloth for not more than 8 seconds. Observe if the markings are completely removed. Compare erased portion of the negative with an unerased portion for evidence of softening or other deleterious effect.

4.3.6 *Permanence.*—Mark a piece of negative several times with the ink and allow to dry. Cut the negative into two parts such that each part will contain about one-half of the markings. Expose one part to the radiation from a Fade-Ometer, Type FDA-R, or its equivalent, for 24 hours. Compare the intensity of the color of the markings with those on the part of the negative which was not exposed. Fold the unexposed negative in such a manner that the fold passes through the markings. Observe if the markings chip or break away from the negative at the line of fold. Markings on negatives when tested for permanence shall remain sufficiently legible that there will be no question regarding the exact identity of the markings.

5. PREPARATION FOR DELIVERY

(For definitions and levels see 6.1.)

5.1 Packaging.—Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*—Unless otherwise specified in the procurement document, 48 bottles of ink in their individual containers shall be packaged in an intermediate container conforming to Federal Specification LLL-B-631 or LLL-B-636. (See 3.11.)

5.1.2 *Level C.*—The ink shall be packaged in intermediate containers according to the manufacturer's usual commercial practice. (See 3.11.)

5.2 Packing.—As far as practicable, exterior containers shall be of uniform size and shape, shall contain identical quantities, and shall be of minimum cube and tare consistent with the protection required. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.*—Ink, packaged in accordance with 5.1.1, shall be overpacked in exterior-type shipping containers conforming to Military Specification JAN-P-108 or Federal Specification PPP-B-585 (class 3), PPP-B-591 (overseas type), PPP-B-601 (overseas type), or PPP-B-621 (class 2). The gross weight of each pack shall be limited to 200 pounds. Containers, except those covered by Military Specification JAN-P-108, shall be provided with a case liner conforming to Military Specification MIL-L-10547 and shall be sealed in accordance with the appendix thereto. As an alternative, Military Specification JAN-P-108 containers may be sealed at all joints and seams with tape conforming to Military Specification JAN-P-127.

5.2.2 *Level B.*—Ink, packaged in accordance with 5.1.1, shall be overpacked in domestic-type exterior containers conforming to Federal Specification LLL-B-631, LLL-B-636, PPP-B-585, PPP-B-591, PPP-B-601, or PPP-B-621, or Military Specification MIL-B-10377. The gross weight of each pack shall be limited to 500 pounds. When fiberboard containers are used, the fiberboard shall meet the special requirements table of Federal Specification LLL-B-631 or LLL-B-636, as applicable.

5.2.3 *Level C.* — Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest applicable rate. Containers shall conform to Consolidated Freight Classification Rules or the regulations of other common carriers as applicable. (See 5.3.)

5.3 Physical protection.—Cushioning and blocking as required shall be in accordance with Military Specification JAN-P-100, ex-

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cept that for domestic shipments waterproofing requirements for cushioning materials and containers shall be waived. The drop tests of Military Specification JAN–P–100 shall be waived for level C packs.

5.4 Marking.

5.4.1 *Interior containers.* — Each bottle, chipboard box, and intermediate container shall be marked with the following data:

Ink, Numbering-Machine, for Photographic Negatives

Federal Specification number

Color

Quantity

Stock number

Order number

Manufacturer's name or trade-mark

"CAUTION: DO NOT FREEZE"

Each bottle and each chipboard box shall be marked also with the following information:

Note: For removing this ink from photographic negatives and numbering machine, use_____*

*Insert name of recommended solvent (common or chemical name, not a trade-mark).

5.4.2 *Exterior containers.*

5.4.2.1 *Civil agencies.* — Unless otherwise specified, each shipping container shall be marked with the name of the commodity, the quantity contained therein as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order. Each shipping container shall be marked: "CAUTION: DO NOT FREEZE".

5.4.2.2 *Military agencies.* — Exterior containers shall be marked in accordance with Military Standard MIL–STD–129. All containers shall also be marked with the following warning: "CAUTION: DO NOT FREEZE".

6. NOTES

6.1 Federal Standard 102, for Preservation, Packaging, and Packing Levels, should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment. (See 2.1.)

6.2 Ordering data.—Purchasers should exercise any desired options offered herein and should specify the packaging and packing level required, as applicable, and number of unit packages to be packaged in each intermediate container, if other than specified in 5.1.1. (See 3.8, 4.1, 5.1, 5.1.1, 5.2, 5.4.2.1, and 6.4.)

6.3 Transportation description.—Transportation description applicable to this item is:

Ink, printing, not otherwise indexed by name.

Carload minimum weight, 36,000 pounds.

Truckload minimum weight, 36,000 pounds.

6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.5 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

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Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
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Copies of this specification may be purchased for 10 cents each.

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